

PID CONTROLLERS USED IN VEX ROBOTICS

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What is PID?

When discussing a Proportional-Integral-Derivative (PID) controller, a form of machine control is meant. Machine control refers to the process of operating a system to reach a desired goal and can be as simple as making a motor spin or as complex as completing a spacecraft journey. In this paper, the focus will be on simple machine control systems using only one actuator and sensor as shown on the figure below.

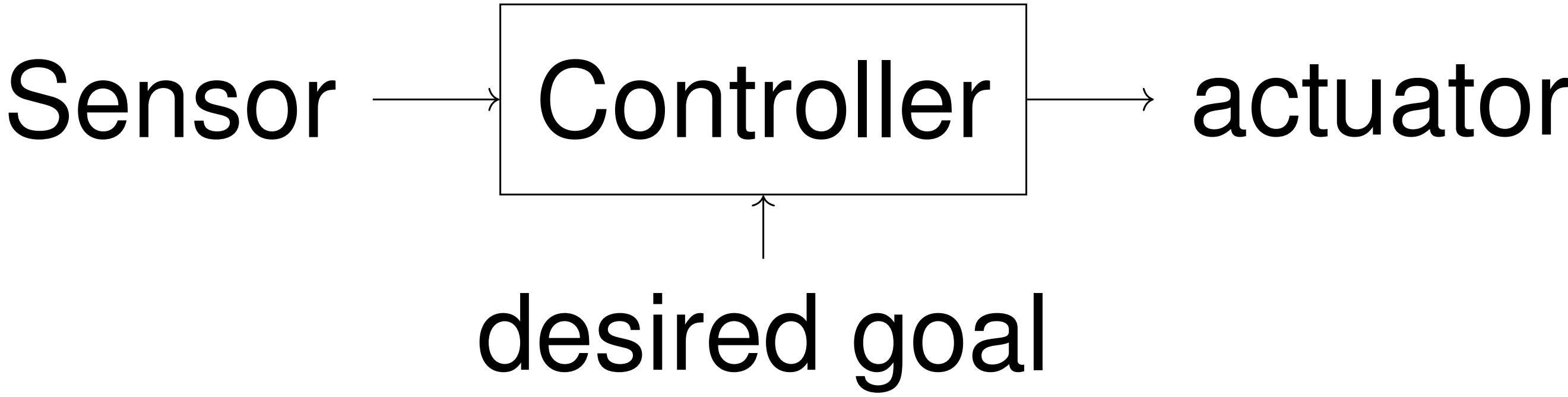


Figure 1: Basic structure of a machine control system. (own image)

A PID controller (u) computes a correcting signal by combining three different terms, each multiplied by a respective constant (K) as shown in the following equation:

$$u(t) = K_P \cdot P(t) + K_I \cdot I(t) + K_D \cdot D(t)$$

The three terms are the proportional change (P), integral (I) and derivative (D) of a calculated error (e). This error (e) is calculated by subtracting a given value the system is supposed to reach with the current measured value.

$$e(t) = \text{sensorValue}(t) - \text{requestedSetpoint}$$

The constants K_P , K_I and K_D are different in each system and are acquired through PID tuning.

Proportional

The proportional term $P(t)$ within a PID system refers to the difference between the current value and the requested setpoint.

$$P(t) = e(t)$$

- + Easy to implement
- + robust
- Possible steady-state-error

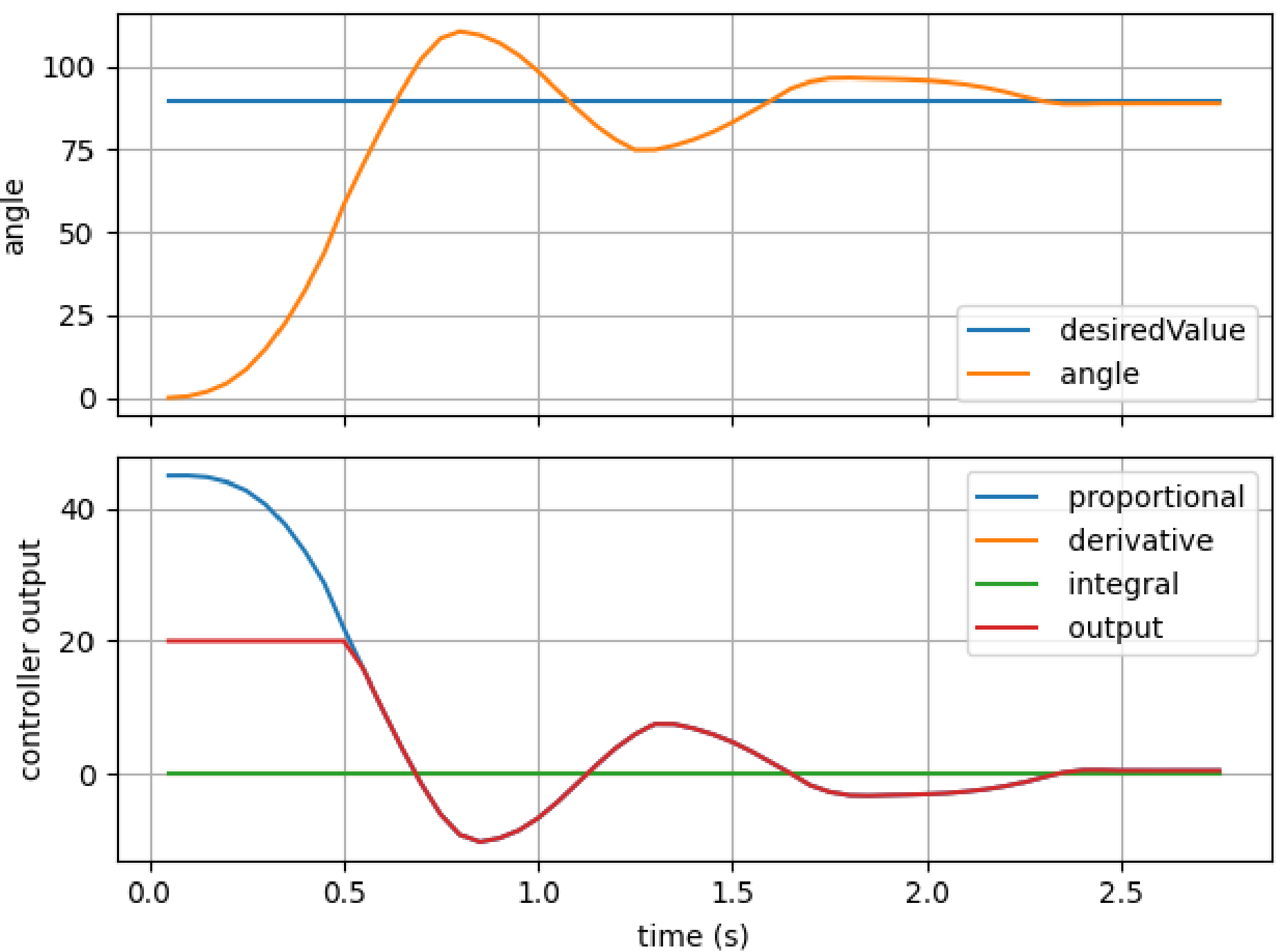


Figure 2: Graph showing a self-tuned P controller

Integral

The integral term $I(t)$ within a PID system is the integral of the error values over time. The term accumulates over past errors.

$$I(t) = \int e(t)dt$$

- + effective to fight steady-state-error
- prone to windup
- can cause oscillations if set high
- can saturate the controller

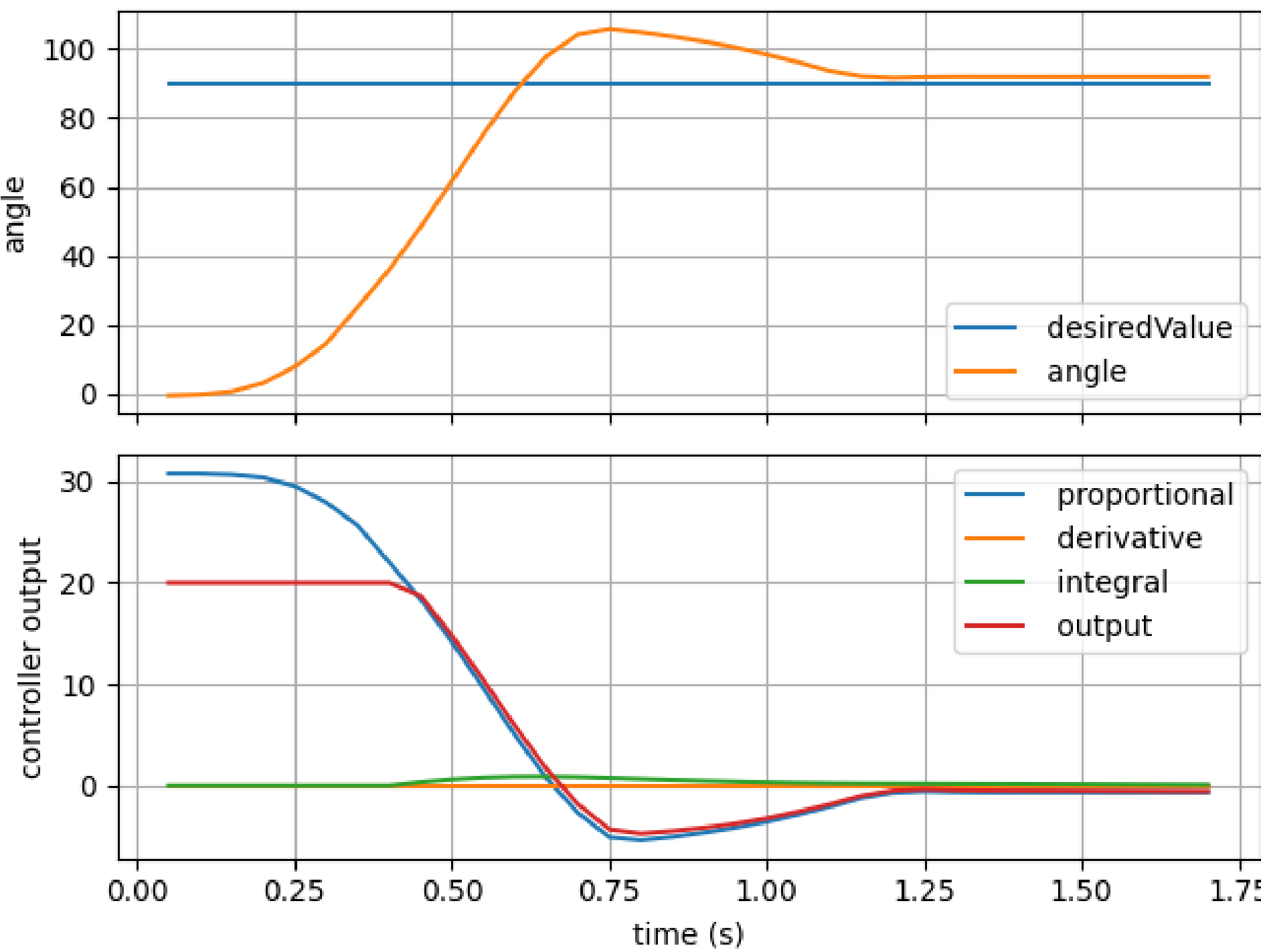


Figure 3: Graph showing a self-tuned PI controller

Derivative

The derivative term $D(t)$ within a PID system is the derivative of the error over time. This means that the derivative term adds the change of error to the PID equation.

$$D(t) = \frac{de(t)}{dt}$$

- only reacts to other terms
- + dampens other terms
- + removes heavy oscillations
- prone to signal noise

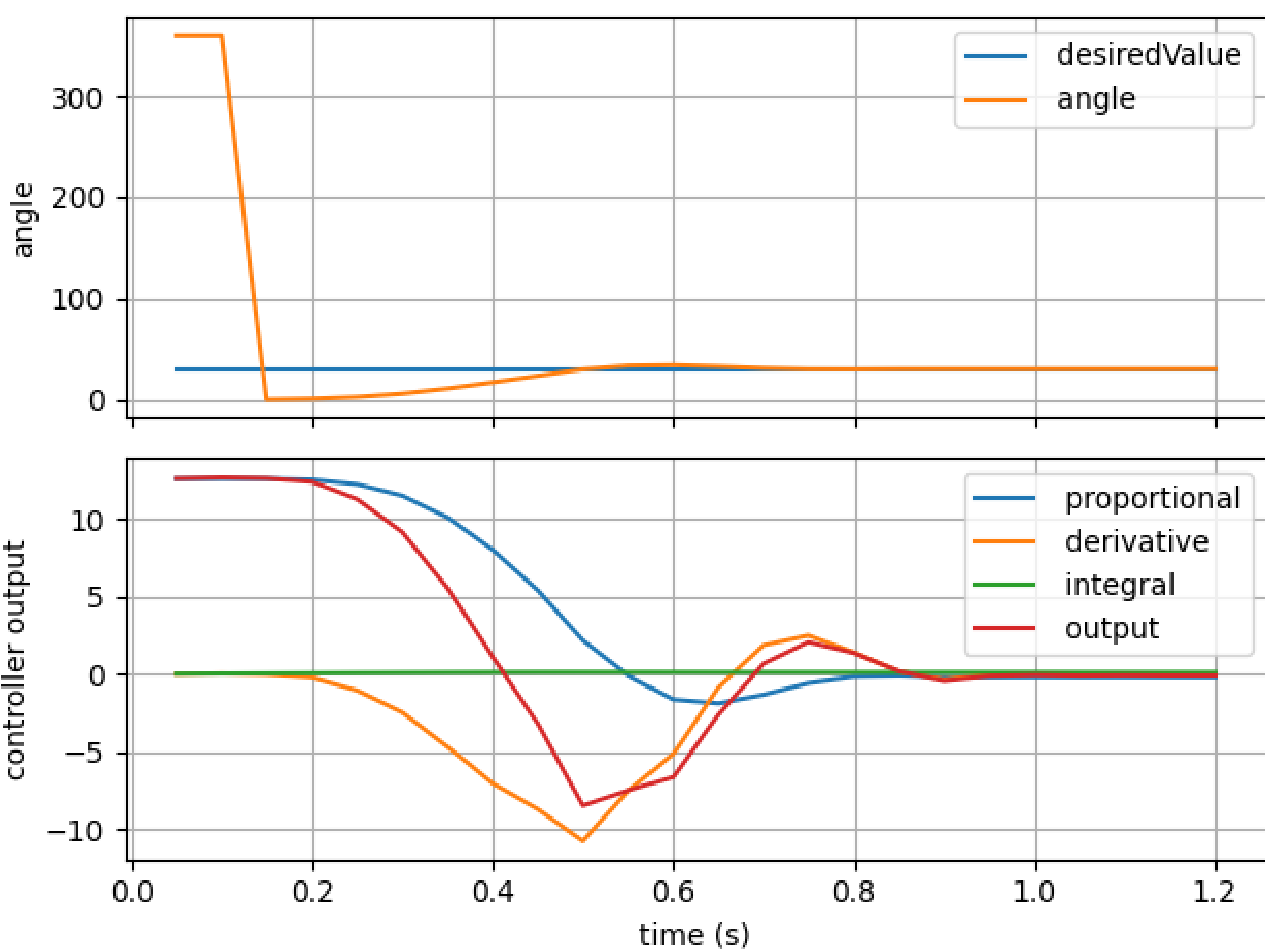


Figure 4: Graph showing a self-tuned PD controller

PID CONTROLLER

